

Wk9 Lab Session

Laura Sun

2025-11-30

Section 1: Proportion of G/G in a population

Download a csv from Ensemble < https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39876388-40046389;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read csv file for MXL:

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

```
## Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
## 1 NA19648 (F) A|A ALL, AMR, MXL -
## 2 NA19649 (M) G|G ALL, AMR, MXL -
## 3 NA19651 (F) A|A ALL, AMR, MXL -
## 4 NA19652 (M) G|G ALL, AMR, MXL -
## 5 NA19654 (F) G|G ALL, AMR, MXL -
## 6 NA19655 (M) A|G ALL, AMR, MXL -
## Mother
## 1 -
## 2 -
## 3 -
## 4 -
## 5 -
## 6 -
```

```
table(mxl$Genotype..forward.strand.)
```

```
##
## A|A A|G G|A G|G
## 22 21 12 9
```

```
table(mxl$Genotype..forward.strand.)/nrow(mxl)*100
```

```
##
## A|A A|G G|A G|G
## 34.3750 32.8125 18.7500 14.0625
```

Let's look at a different population:

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

```
table(gbr$Genotype..forward.strand.)
```

```
##
## A|A A|G G|A G|G
## 23 17 24 27
```

```
round(table(gbr$Genotype..forward.strand.)/nrow(gbr) * 100,2)
```

```
##  
##   A|A   A|G   G|A   G|G  
## 25.27 18.68 26.37 29.67
```

This variant that is associated with child asthma is more frequent in the GBR population than MKL population.

Let's dig further:

Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

How many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")  
head(expr)
```

```
##   sample geno      exp  
## 1 HG00367  A/G 28.96038  
## 2 NA20768  A/G 20.24449  
## 3 HG00361  A/A 31.32628  
## 4 HG00135  A/A 34.11169  
## 5 NA18870  G/G 18.25141  
## 6 NA11993  A/A 32.89721
```

```
nrow(expr)
```

```
## [1] 462
```

Sample size of each genotype:

```
table(expr$geno)
```

```
##  
## A/A A/G G/G  
## 108 233 121
```

Median expression levels of each genotype:

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
expr %>% group_by(geno) %>% summarize(median=median(exp))
```

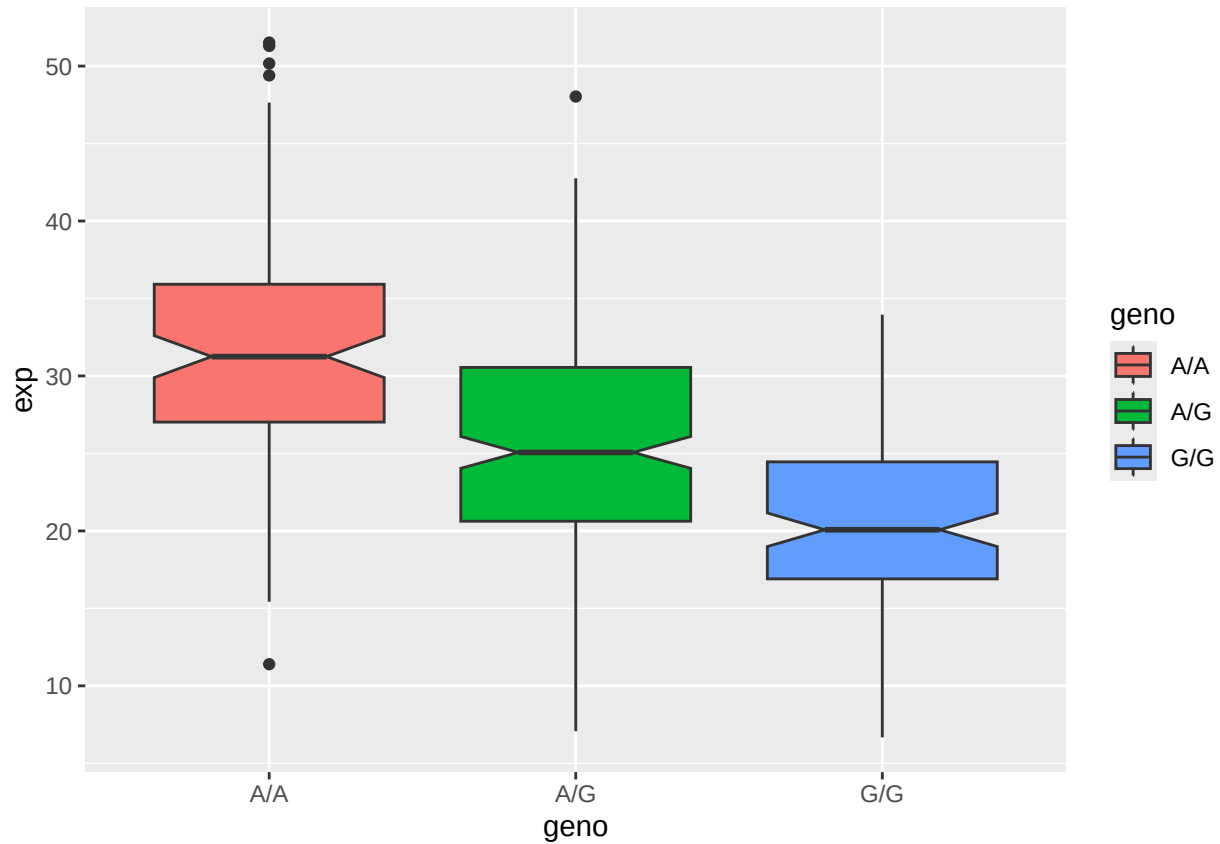
```
## # A tibble: 3 x 2  
##   geno median  
##   <chr>   <dbl>
```

```
## 1 A/A      31.2
## 2 A/G      25.1
## 3 G/G      20.1
```

```
library(ggplot2)
```

Let's make a boxplot:

```
ggplot(expr) + aes(geno, exp, fill=geno) + geom_boxplot(notch=TRUE)
```



From this plot, we see that the relative expression levels are $A/A > A/G > G/G$, with a relatively big difference. SNP does effect the expression of ORM DL3.